

# EduVision\_DV Dashboard Testing Report

## 1. Objective

Validate KPI calculations, verify ranking calculations, test dashboard interactions, validate educational metrics, and record known limitations before final submission.

## 2. Test Basis

The final Excel analytical model was independently inspected. The four final dashboard screenshots were used as visual evidence. KPI card values were recomputed from the final Excel sheets and compared with the displayed values.

## 3. Data Integrity Results

dim\_university: 4,708 unique university IDs; duplicate IDs = 0 — PASS.

dim\_country: 136 unique country IDs; duplicate IDs = 0 — PASS.

Fact composite key university\_id + year + source: 0 duplicates — PASS.

Invalid university foreign keys: 0 — PASS.

Invalid non-null country foreign keys: 0 — PASS.

Country education records: 119 countries represented — PASS WITH NOTE.

## 4. KPI Validation Results

The displayed KPI cards are consistent with independent calculations from the final Excel model after Tableau rounding. Examples: Overall Score 52.50 vs independent average 52.5008; Academic Reputation 20.29 vs 20.2867; Research Impact 37.22 vs 37.2192; Faculty Student Ratio 19.07 vs 19.0717; International Students 9.95% vs 9.9487%; Total Students 92.41M vs 92,405,187; Research Score 23.38 vs 23.3824; Citation Score 42.98 vs 42.9805; Research Productivity 50.03 vs 50.0282; Total Countries 136; Education Countries 119; Average Literacy 87.97 vs 87.9667; Average Tertiary Enrollment 44.33 vs 44.3255.

## 5. Dashboard Functional Testing

PASS — University Overview opens and renders correctly.

PASS — Research Analytics opens and renders correctly.

PASS — Student Analytics opens and renders correctly.

PASS — Country Comparison opens and renders correctly.

PASS — Left-side navigation was manually checked.

PASS — Year filter was manually checked.

PASS — University/Country filters were manually checked.

PASS — Dashboard linking was manually checked.

## 6. Ranking Validation

The University Overview ranking KPI was checked against the final performance fact table. The minimum global rank is 1, matching the displayed 1.000.

## 7. Educational Metric Validation

Country-level education metrics were checked against fact\_country\_education. The final model contains 119 countries with education records. Average adult literacy is 87.9667% and average tertiary enrollment is 44.3255%, matching the displayed 87.97 and 44.33.

## 8. Final QA Conclusion

The final analytical model is internally consistent for identifiers, fact-table composite keys, referential integrity, and tested KPI calculations. No major broken visualization is visible in the supplied final dashboard screenshots. The four dashboard interaction tests have also been manually verified as working.